CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

Apr-May 2018

EC245/207.01/207 Control Systems

Date: 07.05.2018, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 The system with unity feedback having transfer function **[07]**

$$G(s)=\frac{128}{s(s^7+3s^6+10s^5+24s^4+48s^3+96s^2+128s+192)}$$

Find total number of poles on left hand of s-plane also comment on stability.

Q - 2 **Answer any two questions.** **[14]**

I. A feedback Control System has an open loop transfer function

$$G(s)H(s)=\frac{K}{s(s+3)(s^2+2s+2)}$$

Sketch the Root Locus as K is varied from 0 to ∞.

II. Consider a unity feedback control system with an open loop transfer function

$$G(s)H(s)=\frac{K(s+1)(S+2)}{(s+0.1)(s-1)}$$

Draw the root locus of the system with the gain K as a variable.

III. A unity feedback control system has a n open –loop transfer function of

$$G(s)=\frac{K\left(s+\frac{4}{3}\right)}{s^2(s+12)}$$

Draw the Root Locus.

Q - 3 (a) Define the following terms. **[04]**

1) Transient Response
2) Damping Frequency
3) Steady State Error
4) Rise Time

Q - 3 (b) For a system $G(s)H(s) = \frac{K}{s^2(s+2)(s+3)}$, find the value of K to limit steady state error to 10 when input to system is $1+10t+\frac{40}{2}t^2$. **[05]**

Q - 3 (c) A second order system has unity feedback and open loop transfer function **[05]**

$$G(s) = \frac{500}{s(s+15)}$$

1) What is characteristics equation?
2) What is Natural frequency of oscillations and damping ratio?
3) Calculate Peak time and Peak overshoot.
4) If input is ramp of 0.5 rad/sec, calculate steady state error.

OR

Q - 3 (a) **Do as directed.** **[04]**

1) Derivation of Peak Time
2) Derivation of Peak Overshoot

Q - 3 (b) A unity feedback system has **[05]**

$$G(s) = \frac{40(s+2)}{s(s+1)(s+4)}$$

Determine

1) Type of system
2) All error coefficients
3) Error for ramp input with magnitude 4.

Q - 3 (c) A unity feedback system has an open loop transfer function given below: **[05]**

$$G(s) = \frac{(1+0.4s)}{s(s+0.6)}$$

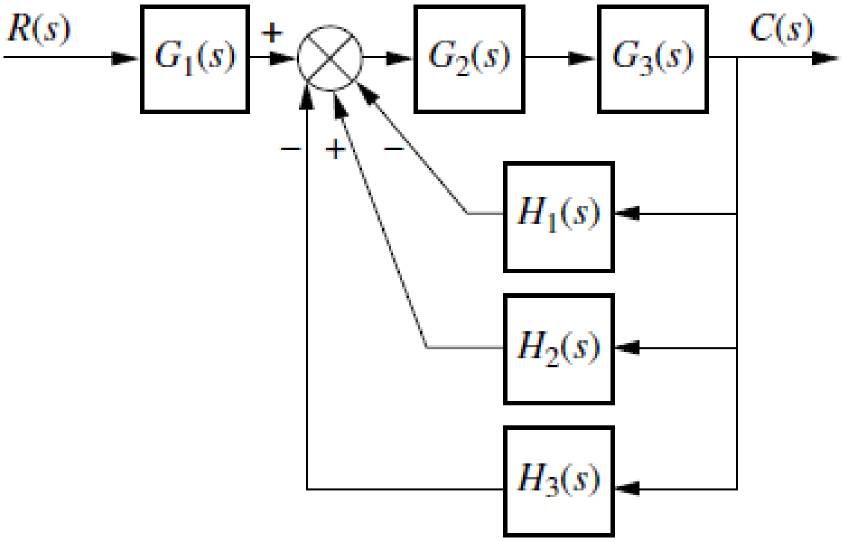
1) Obtain Unit step response.
2) What is Rise time?
3) What is peak overshoot?

SECTION – II

Q-4 (a) Give the differences between open loop control system and closed loop control systems in detail. **[05]**

Q-4 (b) Explain the importance of control system with any real world example. **[02]**

Q-5 (a) Reduce block diagram in canonical form of a system given below. **[04]**

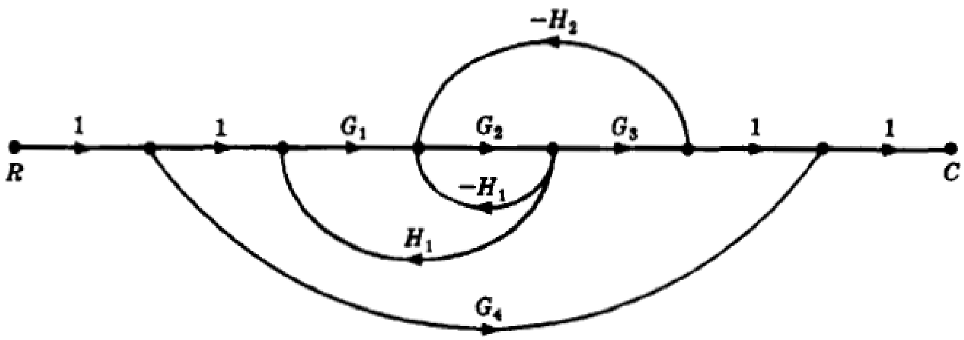


Q-5 (b) **Answer any two questions.** **[10]**

(i) **I.** Using Nyquist criterion, examine the closed loop stability of a system whose open loop transfer function is

$$G(s) = \frac{90}{s^2 + 9s + 18}$$

II. Obtain transfer function from the signal flow graph shown in following figure (Use Mason's gain formula).



III. Find the stability of the following system using Nyquist Criteria.

$$G(s) = \frac{90}{s^2 - 3s - 18}$$

Q - 6 (a) A unity feedback system has an open loop transfer function of **[07]**

$$G(s) = \frac{100}{s(s+1)(s+2)}$$

Draw the bode plots and determine

1) Gain margin
2) Phase margin

3) Gain crossover frequency
4) Phase crossover frequency
5) State about the stability

OR

Q - 6 (a) Sketch the bode plot for the given transfer function, **[07]**

$$G(s)H(s) = \frac{48(s+10)}{s(s+20)(s^2+2.4s+16)}$$

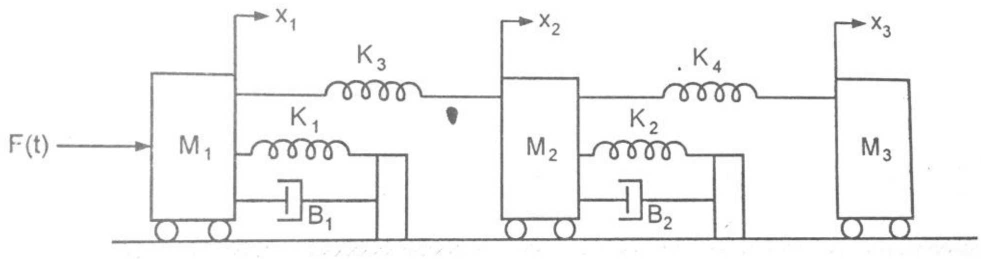
Define the system stability.

Q - 6 (b) Discuss the effect on the system performance: **[07]**

1) Proportional control
2) Integral control
3) Derivative control.

OR

Q - 6 (b) Draw the mechanical network for the mechanical system shown in figure below. Obtain an analogues electrical circuit based on Force- Current analogy. **[07]**



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